higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL SYSTEMS AND CONSTRUCTION
NQF LEVEL 4

22 NOVEMBER 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1
- Natural gas
- Hydroelectric power
- Wind power
- Nuclear power
- Solar power
- Biomass (Any 3 × 1) (3)

1.2 1.2.1 Isolates and protects✓ electrical equipment✓
1.2.2 Removes particles✓ from a flowing gas✓
(2 × 2) (4)

1.3
- Equal emfs
- Same frequency
- In phase
- Same angular displacement (Any 3 × 2) (6)

1.4
- Both line and phase voltages can be obtained
- Earthing is easily obtained (2 × 2) (4)

1.5
- Line to ground
- Short circuit
- Open circuit (3)

[20]

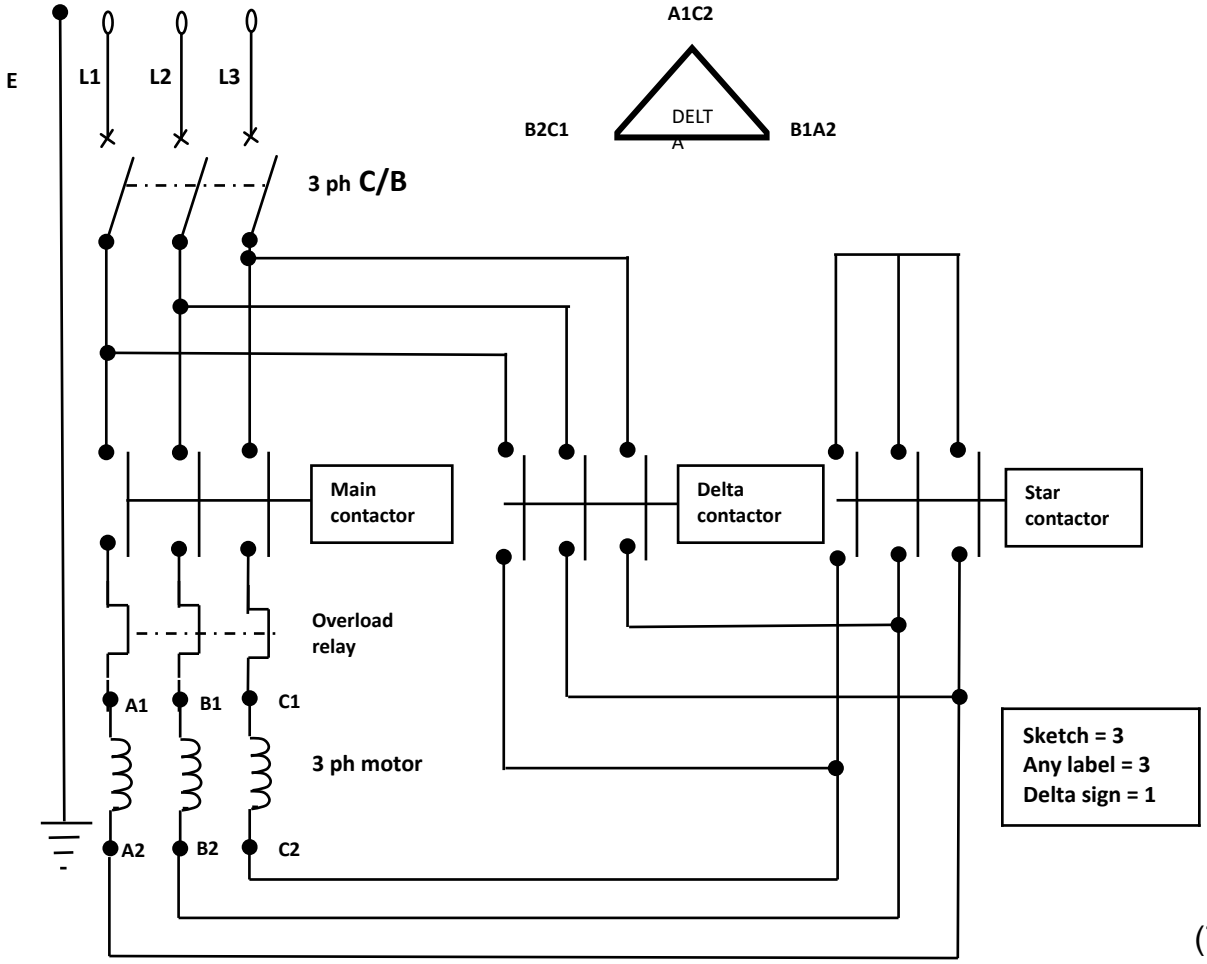
QUESTION 2

2.1 *A solenoid is a magnetic coil that is energised or de-energised✓ to make or break a contacts in a relay.✓*
Or
A solenoid is a magnetic coil in a relay which, when energised, opens or closes contscts in the relay *(2)*

2.2 2.2.1 Protect motors or any other device and associated cables✓ against overload currents✓

2.2.2 Initiate a control sequence✓ to prevent overrunning✓
(2 × 2) (4)

2.3



(7)

2.4 A thumper applies a very brief (millisecond) high-voltage pulse to a cable to determine the exact location of cable failure. (2)

[15]

QUESTION 3

3.1
- Whether the land is flat
- Whether the soil type is suitable
- Whether the route has to curve/change direction
- Whether there are areas that might be flooded
- Suitable position for poles
- Servitudes
- Keeping the cost of construction low
- Environmental conditions (Any 3 × 1) (3)

3.2 Attach one end to the pole, below the cross-arm,✓ and the other end to the anchor rod in the ground.✓ (2)

3.3 A risk assessment is an examination of factors that can have undesirable consequences. It prevents an accident before it can occur. (2)

3.4
- Position of roads
- Building structures
- Variations in the size of lots
- Whether the area is rural or urban (Any 3 × 1) (3)

3.5
- Voltage tests
- Insulation-resistance tests
- Earth-continuity tests
- Earth-leakage tests
- Polarity tests
- Earth-resistivity tests (Any 4 × 1) (4)

3.6 Aluminium conductor steel reinforced (ACSR) (1)
[15]

QUESTION 4

4.1
4.1.1 L
4.1.2 H
4.1.3 N
4.1.4 A
4.1.5 K
4.1.6 D
4.1.7 C
4.1.8 B
4.1.9 E
4.1.10 F
(10 × 1) (10)

4.2
- Connect the maximum available load.✓
- Measure the voltage at the distribution board or point of control.✓
- The acceptable voltage should be within the regulatory limit of ±5% of the supply voltage.✓✓ (4)

4.3 To prevent shock or death✓ as there may still be energy in the reactive component, even if the circuit is switched off✓ (2)

4.4
- SANS 10142
- Occupational Health and Safety
- Driven Machinery Regulations
- Construction Regulations
- Electrical Installations Regulations (Any 2 × 1) (2)

4.5 Some parts of the installation not covered by the certificate can be documented in the comments section. (2)
[20]

QUESTION 5

5.1 A Buchholtz relay on a transformer will detect an abnormal condition✓ within the tank and send an alarm or a trip signal.✓ (2)

5.2 •Windings
•Bearings (2)

5.3

CAUSES	CORRECTIVE ACTION
•Oil level could be low •Tripping value could be set lower	•Fill up the oil •Adjust tripping value to correct trip current

(2 + 2) (4)

5.4 •Divide and conquer
•Fault indicators
•Thumper test
•Burner technique
• Earth-gradient detection procedure (Any 3 × 1) (3)

5.5 • Connect the power supply on the primary terminal of the transformer.
• Measure the voltage at the primary terminals.
• Measure the voltage at the secondary terminals.
• The side that is open circuited or faulty, will not read any voltage. (4)

[15]

QUESTION 6

6.1 It is a system where the user is connected separately✓ to both a renewable energy source✓ and the power grid.✓ (3)

6.2 Total power consumption (in kWh/day):

$$P_{TOTAL} = 5{,}04 + 0{,}12$$
$$P_{TOTAL} = 5{,}052 \text{ kWh/day} ✓✓$$

His expenditure per month:

$$COST_{TOTAL\ PER\ MONTH} = POWER - CONSUMPTION_{TOTAL} \times DAYS \times COST_{PER\ KILOWAT}$$
$$COST_{TOTAL\ PER\ MONTH} = 5{,}052 \times 30 \times 1{,}25$$
$$COST_{TOTAL\ PER\ MONTH} = R189{,}45 ✓✓$$ (4)

6.3

$$E_{MIN} = Ah \times Battery\ (voltage)$$
$$E_{MIN} = 2 \times 24$$
$$E_{MIN} = 48\ kWh$$ ✓✓

$$Days = \frac{E_{MIN} \times BDD}{E_{DAILY}}$$
$$Days = \frac{48 \times 0,7}{1,6}$$
$$Days = 21\ days$$
✓✓ (4)

6.4
- Do measurements.
- Collect photographs.
- Identify potential safety hazards (Any 2 × 1) (2)

6.5 It regulates the current from the solar panels✓ to prevent the batteries from overcharging.✓ (2)

[15]

TOTAL: 100

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